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Below is the *Précis* of: Jonathan Birch’s [The edge of sentience: Risk and precaution in humans, other animals, and AI](#). Oxford Academic.

The entire book is openly accessible at [this link](#). It is the book itself that is to be reviewed. As there will be multiple book-reviews, authors need only review the portions of the book that are relevant to their own field of expertise and interest.

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Précis of *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*

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Abstract: We often face grave practical decisions that seem to hinge on whether a system is sentient. This family of cases includes invertebrate animals, people who are unresponsive after brain injury, fetuses, neural organoids, and now AI technologies. We must decide what to do despite ongoing disagreement about the nature of sentience. In our state of uncertainty, we should pragmatically transform the question from “Is it sentient?” to “Is it a sentience candidate, an investigation priority, or neither?”. When a system is a sentience candidate, it is negligent to fail to consider precautions. We should instead evaluate precautions for their proportionality using the “PARC tests”. When we think about the problems in this way, we see that overconfidence about the absence of sentience has repeatedly led decision makers to neglect serious risks. Erring on the side of caution requires many revisions to current practice in many areas of human activity.



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1. Introduction. In cases at the edge of sentience, grave practical decisions seem to hinge on whether sentience is present or absent in a person, animal, or other cognitive system. We face this type of predicament in relation to invertebrate animals, people who are unresponsive after brain injury, fetuses, neural organoids, and now AI technologies that reproduce brain functions and/or mimic human behavior. Because we are unsure whether sentience is there, we are unsure what to do. We need frameworks for helping us manage that uncertainty and reach decisions.

It is extremely important, in my view, to discuss the cases together. I find that comparative questions come up all the time. *How can you advocate for stronger protections for animals but not for human fetuses? Why do you see an insect as having greater moral standing than a chatbot? Why do you support replacing animal research with organoid research if the organoids themselves may be sentient?* If we don't have a common framework for handling the various cases, we will be left unable to answer questions like these.

My Open Access book *The Edge of Sentience: Risk and Precaution in Humans, Other Animals and AI* (Birch 2024) develops a consistent precautionary framework for the whole family of cases. It rests on the insights that we must *err on the side of caution* in these cases, take *proportionate* steps to manage risk, and avoid *reckless or negligent* behaviour. Where sentience is in doubt, we should give these systems the benefit of the doubt as far as we reasonably can. What that means in practice varies from case to case.

This article is a précis of the book. It will feel very quick and a little dense! Please consult the book at any points where further detail seems needed or where you want things to move more slowly. I've also made a few simplifications. If anything feels oversimplified, please check the book to see if it gets a more careful treatment. I hope the précis can serve as a menu that helps you find the sections of the book most relevant to you. In the AI section, I have updated the discussion slightly, since 2024 is already a long time ago in the AI world.

2. The concept of sentience. Sentience has broader and narrower senses. In a broad sense, it refers to any capacity for feeling. Philosophers call this a capacity for “phenomenal consciousness”, where a state is phenomenally conscious if there is “something it's like” to be in that state (Nagel 1974). In a narrower sense, sentience is specifically the capacity to have *valenced* experiences—experiences that *feel bad or feel good to the subject*—such as experiences of pain and pleasure. I use this narrower sense of the term as a working definition.

Defining sentience as the capacity for valenced experience can be common ground, despite continuing disagreement about the nature of phenomenal consciousness and the nature of valence. We can hope that the “something it's like” definition of phenomenal consciousness will eventually be superseded by a more informative successor, but this must await much more theoretical consensus than currently exists. There are, at present, many theories of phenomenal consciousness, displaying massive variation (Seth and Bayne 2022).

Valence, meanwhile, is one of the two major dimensions, along with arousal, that characterize affective states, and it is the dimension that matters most from an ethical point of view. To see why valence matters more than arousal, imagine a drug that flips the valence of pain without altering the arousal aspect: a high-arousal pain state becomes a high-arousal pleasure state. Now imagine a second drug that alters arousal while leaving valence unchanged: the pain still feels extremely bad, but it no longer rouses you from drowsy lethargy. You'd surely take the first drug! The second might have some value, but the first, if it existed, would transform our lives vastly for the better.

Why do valenced experiences matter so much? Three major ethical theories give different answers. For classical utilitarians such as Peter Singer (2024), ethics is about calculations that add up the pains and pleasures caused by an action, and valenced experiences are what count in these calculations. For Christine Korsgaard (2018), valenced experience involves a kind of implicit valuation: what feels good is represented as good for me, whereas what feels bad is represented as bad for me. We necessarily take our own implicit valuations to be tracking facts about what is objectively good and bad (my pain is *objectively* bad, not just bad for me), and the pressure of consistency compels us to say the same about all sentient beings. For Martha Nussbaum (2022), valenced experience implies “a felt orientation towards what is seen as good and a felt aversion to what is seen as bad”, and only beings with this orientation can be wronged.

Valenced experiences do not have to fall under traditional human categories. There may well be valenced experiences in other beings that our everyday categories (such as pain, anxiety, fear, anger) fail to capture well. Think, for example, of the Guiana dolphin, *Sotalia guianensis*, which possesses electroreceptors that register electric fields (Czech-Damal et al. 2011). Perhaps, for these dolphins, there is a feeling of “unpleasant disorientation of one’s electroreceptors” that is very different from any human feeling and has no unique word in any human language. We should expect there to be far more valenced experiences in Heaven and Earth than are dreamt of in our ordinary emotional vocabulary.

3. The zone of reasonable disagreement: philosophy of mind. People disagree constantly about sentience. The disagreement is often reasonable, because current evidence leaves much uncertain. It is possible to characterize a “zone of reasonable disagreement” that includes a lot, while still excluding views that rely on unsubstantiated empirical claims, dogmatically cling to refuted hypotheses, and/or advance morally abhorrent (e.g. sadistic) positions.

I spend three chapters (Chapters 3-5) mapping this zone. My aim is not to give an *overview* of all the topics on which people currently disagree. That would require many books. Instead, my aim is to capture the *sources* of persistent disagreement: sources we should expect to keep throwing up new disagreements, as a volcano expels fresh lava, long after this generation’s disagreements cool and solidify. (**Figure 1**).

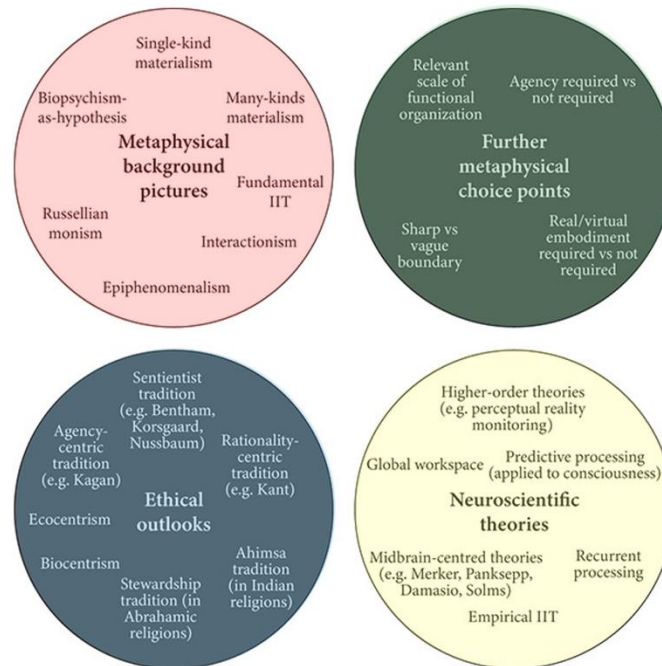


Figure 1: A map of the major landmarks in the zone of reasonable disagreement.

Some thorny sources of disagreement have their origins in the philosophy of mind. One delicate issue is how to handle “epiphenomenalism”, the idea that sentience makes *no functional difference whatsoever* to cognition or behaviour. This view has had serious defenders, notably T. H. Huxley (1874), and it can’t be completely ruled out. Some say: if you can’t even rule *that* out, how can you study sentience scientifically? However, we don’t need to conclusively rule out epiphenomenalism to regard it as empirically unlikely. Sentience bears the hallmarks of being an evolved adaptation, and adaptations must have effects (James 1890/1918; Klein 2019). While it’s true that the science of sentience is premised on the assumption that sentience makes a difference to cognition and behaviour, there is nothing wrong with premising a research program on an empirically plausible assumption.

Another source of disagreement is the age-old clash between materialism and dualism. Panpsychism, the view that all matter is sentient, lurks in the shadows, offering a third way when the traditional options start to feel frustrating. Materialism is nowadays the mainstream position, but materialists still disagree enormously among themselves. A significant faultline is between materialists who regard sentience as a real, objective mechanism or process in the brain and those who suspect the word merely gestures loosely towards a multitude of mechanisms or processes. I call the second group the *many-kinds materialists*, though some of them prefer the label “illusionists” (Frankish 2016; Dennett 2017). My sympathies lie with the first group. If the second group turns out to be right, then we will eventually need to replace “sentience” with a multitude of more precise successor concepts.

Meanwhile, dualism and panpsychism have the status of radical alternatives. Do they have a place in the zone of reasonable disagreement? I say: *potentially, yes*, provided they are cautiously formulated. For example, a “panprotopsychism” (Chalmers 2017) that posits *precursors* to sentience in fundamental particles might be reasonable, whereas a version that attributes a complex, evolved adaptation to fundamental particles is not. I take a similar approach towards “biopsychism”, the view that all life is sentient. It’s fair to offer it as a speculation but incorrect to claim it to be empirically supported (on this, see my replies to Arthur Reber and colleagues in this journal; Birch 2017; Crump et al. 2022b).

There are several other axes of philosophical disagreement that lead people to disagree about the edge of sentience. One concerns the importance of agency and embodiment, real or virtual. Another concerns the scale of functional organization that matters. Can small-scale (e.g. within neuron) functions make a difference to the presence or absence of sentience, or is large-scale functional organization all that counts? A third concerns whether the edge of sentience is sharp or blurred: must there be a sharp transition between the absence of sentience and its presence in its most minimal forms, or can there (as with baldness, tallness, oldness) be borderline cases in which there is no determinate fact of the matter one way or the other? When thinking about particular cases, we should look for ways forward that don’t rely on controversial assumptions about these issues.

4. The zone of reasonable disagreement: ethics. That same message carries over to ethical disagreement. If we understand the shape of the landscape, we can start to look for “Goldilocks” proposals that will be acceptable to all sides.

Here the major divide is between sentientist ethical outlooks, which take sentience as necessary and sufficient for moral standing, and alternatives that deny the necessity and/or sufficiency claim. I am sympathetic to sentientism but I take the alternatives seriously too, and I want to bring those other groups with me.

Among the alternatives are biocentrist or ecocentrist views that place fundamental ethical value on life or nature (respectively). However, the difference with sentientism is often exaggerated. Reasonable versions of bio/ecocentrism still grant major significance to sentience. A version that denied any major significance to sentience would be unable to justify prioritizing sentient animals over bacteria when their interests conflict, but no one has seriously claimed that we should not (for instance) prioritize a dog over the bacteria that have infected it.

Agency-centric and rationality-centric positions provide further alternatives. An example of the latter is orthodox Kantianism, which allows only indirect duties (formally owed to ourselves) concerning non-rational sentient beings. We owe it to ourselves, on this view, to cultivate a sense of empathy, and that is why we should be sensitive to the suffering of other animals (Kant 1797/2017). Even though I personally think this view gets things backwards, I have to acknowledge it as internally coherent.

Another challenge comes from the Abrahamic religions, which give very limited moral standing to non-human sentient beings. We can, however, find in all of them support for

duties of stewardship, including the duty to avoid causing gratuitous suffering. Hinduism, Buddhism and Jainism, meanwhile, share important elements of the sentientist view.

5. The zone of reasonable disagreement: science. To have a science of sentience at all, we need reliable ways of disentangling felt and unfelt processing. Yet here too, we find entrenched disagreements. In the case of vision, long-running debates about “blindsight” epitomize the main sources of uncertainty (Phillips 2021).

In the absence of secure ways of dissociating sentience from unfelt processing, we lack certainty even on what is perhaps the most basic question one can ask about sentience in the mammalian brain: *is it primarily subcortical or primarily cortical?* On the first picture, sentience wells up directly from subcortical mechanisms without the need for any further cortical processing (Panksepp 1998 is a clear example). On the second, subcortical circuits produce coordinated behavioural responses without the involvement of any kind of sentience, with sentience requiring the cortical “re-representation” of subcortical activity (LeDoux 2021 is a clear example). Unfortunately, current evidence does not allow us to choose confidently between these pictures.

The research programs that led to these pictures have had a strong mammalian focus, with most of the evidence coming from humans, macaques, and rats. They have not yielded theories that apply in a clear, unambiguous way to non-mammalian animals. In these cases, we are left with a good amount of behavioural evidence, and some understanding of the underlying neural mechanisms, but a lack of mature theories as to how similar the neural mechanisms must be to their mammalian homologues/analagues to support sentience. There have been some valuable attempts to fill that gap (e.g. Feinberg and Mallatt 2016; Ginburg and Jablonka 2019) but we must acknowledge it as an area of major uncertainty.

6. Converging on precautions. These disagreements run deep, and they will be with us for the long term. Can we nonetheless find points of consensus that reach right across the zone of reasonable disagreement? I say: *yes, we can*. To do this, we need two bridging concepts: *sentience candidates* and *investigation priorities*.

A system is a *sentience candidate* if there is an evidence base that (a) implies a realistic possibility of sentience that it would be irresponsible to ignore when making policy decisions, and (b) is rich enough to allow the identification of welfare risks and the design and assessment of precautions.

Meanwhile, a system is an *investigation priority* if it falls short of these conditions and yet (a) further investigation could plausibly lead to its recognition as a sentience candidate and (b) it is affected by human activity in ways that may call for precautions if it were a sentience candidate.

The pivotal claim of the book is this: we should make a pragmatic shift from the question “Is it sentient?” to “Is it a sentience candidate, an investigation priority, or neither?”. The first question is short-term intractable for all the reasons noted earlier, whereas the second question is short-term tractable.

By transforming the question in this way, we turn it into a question that is partly scientific, partly ethical. We cannot just review the evidence and say “well, that settles it”. We must also reach a judgement about whether the evidence implies risks that it would be *irresponsible to ignore*. We cannot shy away from that ethical component of the judgement. It is essential.

A further key claim is that the relatively inclusive theories of Panksepp, Damasio, and Merker, which link sentience to evolutionarily ancient mechanisms in the midbrain, describe realistic possibilities it would be irresponsible to ignore, so a system with these mechanisms is a sentience candidate (**Figure 2**). This becomes very important when considering animals without a neocortex. One might retort: *but that midbrain-centric family of theories only has a 10% chance of being right!* I am myself more sympathetic than this; I would give it about a 50% chance of being right. But even a 10% chance is one that we must take seriously when it leads to serious risks.

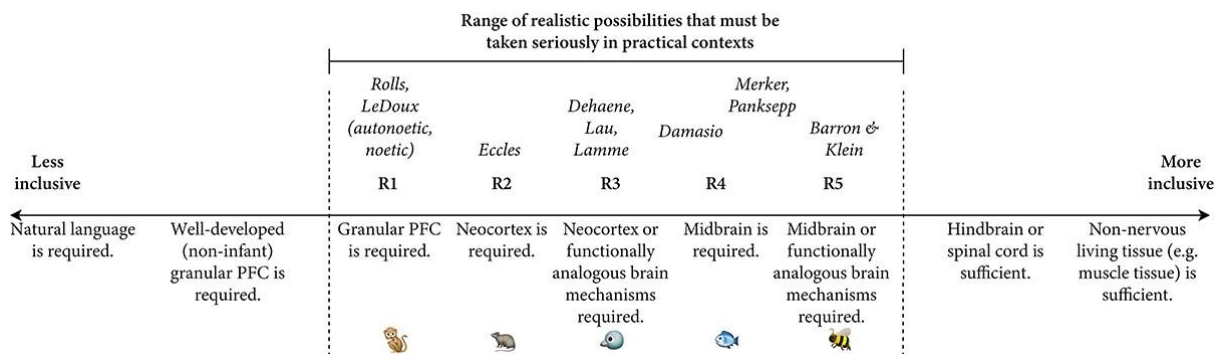


Figure 2: Views about the neural system requirements for sentience. There is no consensus about which position within the reasonable range is correct. The point of agreement is that R1–R5 are all realistic possibilities with reasonable defenders.

What about the ethical part of the zone? Here we find an ethical consensus around a duty to, at minimum, *avoid causing gratuitous suffering* to sentient beings. Suffering is not gratuitous if it occurs in the course of a defensible activity despite proportionate attempts to prevent it. Suffering is gratuitous if the activity is indefensible or the precautions taken fall short of what is proportionate.

These ideas are combined in the principle that *when a being is a sentience candidate, it is negligent to make policy decisions that affect it without carefully considering precautions. The precautions we take should be proportionate to the risks.*

The sentience candidate concept gives us, in effect, a criterion for negligence. Think of someone who drops a lobster into a pan of boiling water without consulting any relevant evidence or guidance. This is negligent because lobsters are sentience candidates. By contrast, it is not negligent to do the same thing to a cabbage, because the cabbage is not a sentience candidate. This is the profoundly important distinction marked by the sentience candidate concept.

This leaves us with the big question of what “proportionality” requires. There will be no one-size-fits-all answer. Every case requires its own debate (see below). All can agree, however, on the unacceptability of a free-for-all in which everyone can treat sentience

candidates however they wish. To avoid this, we need informed and inclusive democratic processes to reach decisions on proportionality, which can then be revisited periodically and whenever significant new evidence emerges.

The framework enshrines what I see as the four pillars of precautionary thinking: we must not let scientific uncertainty paralyse decision-making, precautions should be proportionate to the threat, our procedures for debating precautions should be democratic and inclusive, and the evidential bar for triggering those procedures should be set intentionally low.

7. Involving the public, debating proportionality. When designing precautions, we must avoid a “tyranny of expert values”. This occurs when the values of a clique of expert advisers determines a policy decision without those values being properly scrutinized by a democratic process. Citizens’ assemblies, composed of random samples of the wider population, can be an attractive way to avoid the tyranny of expert values, when carefully designed.

Citizens’ panels can have advantages over elected assemblies and referendums. These advantages are especially clear when an issue generates deep value conflicts, requires sustained attention and regular revisiting, requires consideration of the interests of beings who cannot vote, and when there are reasons to take the issue outside the run of party politics. Questions of proportionality at the edge of sentience have all of these properties.

To be adequately representative, citizens’ panels/assemblies should have at least 150 citizens but should fission into smaller groups of 10–25 for deliberation. They should have the power to co-opt new members to ensure that important but marginalized perspectives are represented in every deliberative group.

Since members of citizens’ panels do not generally have scientific training, careful thought needs to be given to the structure of deliberation, so that they are not forced into a position of arbitrating scientific disagreement. Their focus should be on whether or not a proposed response can be publicly justified as *proportionate* to an identified risk.

What does it mean, though, to judge a precaution as “proportionate”? What are the key tests? I outline a *pragmatic ideal*: a process we could realistically aim to achieve and that would be able to command public confidence. In this process, the panel is presented with a shortlist of feasible options on which stakeholders have been openly consulted. To each policy option, the panel applies four tests in sequence: *permissibility-in-principle*, *adequacy*, *reasonable necessity*, and *consistency*. I call them the “PARC” tests (**Figure 3**). Proposals that fail a test are set aside. Proposals that pass all four tests are judged proportionate.

The four tests create a division of labour between experts and the public. At each stage, the expert advisers provide input regarding the likely consequences of different policy options, but it falls to ordinary citizens to debate the central ethical questions. These questions can be easily understood and do not require arbitrating scientific disagreements.



Figure 3: The PARC tests for proportionality. A proportionate response to a risk will pass all four tests.

From here, the rest of the book aims to drive debate about specific cases. I don't claim to have all the answers, but I can make *proposals* that I think deserve to be on the table and that I think would pass the PARC tests. Readers should not support my proposals just because I find them proportionate. I am not auditioning for the role of "sentience czar". The proposals should be read as *debate starters*: invitations to reflection and deliberation. Do my proposals pass the tests, or do they fall short, in your view? Please tell me!

8. Beyond the so-called "vegetative state". Sometimes a person, after brain injury, displays sleep-wake cycles but has severely impaired, or entirely absent, responses to external stimuli. Traditionally, attempts have been made to distinguish the persistent vegetative state (PVS) from the minimally conscious state (MCS), which is then subdivided into 'MCS plus' and 'MCS minus'. Yet diagnostic procedures for distinguishing PVS from MCS are subject to high error rates and high uncertainty. Moreover, there are cases of patients who, even after being diagnosed as PVS through a very thorough application of behavioural criteria, show signatures of brain activity in response to commands (Claassen et al. 2025). And there is a realistic possibility that midbrain mechanisms suffice for basic sentience even if cortical injury prevents a patient from following spoken commands in any way (Panksepp et al. 2007).

One reaction has been to rename the PVS as "unresponsive wakefulness syndrome" (Laureys et al. 2010). This is an improvement, but because it is just a new name for PVS, it means there is still a clinical category with the stark implication that the patient is "not even minimally conscious". We need to find ways to move beyond that whole approach to patient classification.

In the context of pain management, an "assume sentient" principle is appropriate. Any signs that would be normally interpreted as signs of pain or distress should still be so interpreted in an unresponsive patient. This treatment should not be reserved for a subset of patients, e.g. those thought to be "minimally conscious". In fact, this "assume sentient" principle can be extended, with care, to all clinically important decisions. This

must be understood as a precautionary assumption and not as evidence for or against any particular prognosis.

In some cases, the whole decision-making team, including clinicians and relatives, agrees that sustaining life is not in the patient's best interests. Tragically, standard practice in this type of situation is to end life by withdrawing clinically assisted nutrition and hydration. The patient dies of hunger and thirst over the ensuing days or weeks. The testimony of patient's families makes it plain how unacceptable this is. Comments from relatives (reported by Kitzing and Kitzing 2015) include:

"When they told me that I felt like screaming. I thought, 'you are MAD!' I didn't even contemplate it. I just don't understand how anyone can sanction that law."

"I couldn't believe that that was the method. Everybody I've talked to has said, 'there must be better ways than that in this day and age.' [...] What a stupid, stupid way of doing it."

"I just couldn't bear that he was starving [...] I just thought, what a horrible—what a dreadful way to live your last days. I just thought it was awful."

This method would not be acceptable for any other sentient being. Faster alternatives must be permitted. Part of the problem now is that, in many jurisdictions, clinicians risk prosecution if they administer lethal doses of sedatives or analgesics. In cases where the withdrawal of food and water has already been legally authorized, clinicians should be assured that mercifully speeding up the process will not lead to punishment.

9. Human fetuses. In the 1980s, growing public awareness that anaesthesia was not routinely used on *newborn babies* during surgery led to a successful campaign to change clinical practice, vindicated by subsequent evidence. This shows us the value of taking a precautionary attitude—and the value of involving the public in the discussion. There are times when the medical profession has been led badly astray by overconfidence and groupthink.

More recent evidence suggests we should recognize human fetuses as sentience candidates from the start of the second trimester. The evidence points to maturing midbrain structures, goal-dependent motor control, apparent anticipation of the consequences and risks of specific movements, and emerging nociceptive responses (Ciaunica et al. 2021), together implying a realistic possibility of sentience in the second trimester that it would be irresponsible to ignore (Derbyshire and Bockmann 2020).

What does this mean for abortion? Many people are concerned that recognizing fetal sentience might be incompatible with recognizing a right to abortion. However, the two issues are logically distinct. The appropriate time limit for abortions plausibly depends either on when the fetus becomes a *person* and/or on the strength of a person's right to *bodily autonomy* (Greasley 2017). A fetus may be sentient without being a person, since personhood involves additional features—including mechanisms of memory and planning that create persistence of identity across time—for which there is no evidence in second trimester fetuses. Abortion can be seen as a situation where the interests of a sentience candidate come into grave conflict with those of an adult person, and sometimes it is right for the conflict to be resolved in favour of the person.

There is, however, a need for honest communication of uncertainty with patients. All too often, women are given misleading assurances about the supposedly proven absence of sentience in fetuses, when, in reality, no one can justifiably have such certainty. False certainty serves no one's interests. New norms of communication need to be formulated by appropriately inclusive processes such as citizens' assemblies.

10. Neural organoids. Human neural organoids are models of the human brain constructed from human stem cells. They are showing excellent promise as potential replacements for animal research. But could they themselves develop sentience?

Scepticism about this idea is appropriate when considering current organoids. This is not because of their size, but because current organoids lack a functioning brainstem or anything equivalent to one. There are nonetheless some troubling early warning signs, suggesting that organoid research may create forms of sentient being in the near future. Organoids left to develop for a long time have demonstrated patterns of electrical activity that showed statistically significant similarities to preterm infants (Trujillo et al. 2019). They can spontaneously develop brain structures, such as optic vesicles, without any deliberate steering from researchers. In one remarkable study, a system mounted on electrode array was trained to play the video game Pong, showing measurable improvement across twenty minutes of gameplay (Kagan et al 2021). In another, a complete model of every stage in the human pain pathway, from nociceptor to cerebral cortex, was constructed (Kim et al. 2025). This area is moving fast, to say the least.

Researchers with very different views about the neural basis of sentience should unite behind the 'brainstem rule': if a neural organoid develops or innervates a functioning brainstem (including the midbrain) that regulates arousal and leads to sleep-wake cycles, then it is a sentience candidate. An artificial brainstem substitute may also be enough.

When a system is a sentience candidate, we should discuss proportionate steps to protect its welfare, despite continuing uncertainty and doubt. What steps might be proportionate? If organoid research leads to the creation of new sentience candidates, a moratorium (time-limited ban) or indefinite ban on the creation of this particular type of organoid may be defensible, but bans should avoid indiscriminate targeting of all organoid research. An alternative approach, more consistent with existing approaches to animal research and with the imperative to develop replacements, is to require ethical review and harm-benefit analysis whenever a neural organoid is a sentience candidate.

11. Invertebrate animals. All adult vertebrates, including fishes, are clearly sentience candidates, in the sense that there is *at minimum* a realistic possibility of sentience that it would be irresponsible to ignore. Debate in recent years has rightly shifted towards invertebrates, where sentience candidature is more contestable.

Octopuses are often regarded as sentient due to their large brains and impressive intelligence. They were recognized by the UK's Animals (Scientific Procedures) Act in 1993 for these reasons. Yet *sentience is neither intelligence nor brain size*, and we should be aware of the possibility of decouplings between intelligence, brain size, and sentience in the animal kingdom. Sentient animals might not impress us with their intelligence or brain size. Precautions to safeguard animal welfare should be driven by markers of sentience, not by markers of intelligence, brain volume or neuron count (Shriver 2022).

That said, even when we restrict our focus to markers of sentience, octopuses are clearly sentience candidates. I led a review in 2021 that considered the evidence of sentience in cephalopod molluscs (octopuses, squid, cuttlefish, and nautilus) and decapod crustaceans (including many crabs, lobsters, crayfish, and shrimps). We developed a pluralistic framework based on eight criteria that can be applied to invertebrates and that have an empirically credible theoretical motivation (Birch et al. 2021; Crump et al. 2022a). Our list refines earlier lists by Smith and Boyd (1991), which is in places much too vague, and by Sneddon et al. (2014), which contains some overlap between criteria. We reported with high confidence that octopuses satisfied at least seven of our eight criteria.

More broadly, the evidential picture is complex and gradated. The question of when *enough* evidence has amassed to imply a realistic possibility of sentience that it would be irresponsible to ignore is a difficult judgement call with an ethical component. In my view, the coleoid cephalopods (octopuses, squid, cuttlefish) and decapods of the Pleocyemata suborder (including true crabs, lobsters and crayfish) are clear sentience candidates, whereas decapods of the Dendrobranchiata suborder (including penaeid shrimps) are investigation priorities. There is an urgent need for more research on penaeid shrimps, given the vast scale of shrimp farming. Moreover, some of the practices used in industry (notably eyestalk ablation) are so extreme that the industry can and should move away from them pre-emptively, even while the evidence of sentience is admittedly thin.

What about insects? The idea of pain in insects has often been dismissed on the grounds that they sometimes continue normal feeding and mating behaviours despite catastrophic injury. But this is compatible with insects having aversive experiences in response to other stimuli, such as heat, and there is emerging evidence that they do (Crump et al. 2023). Moreover, there is evidence that insects have a behavioural core control unit functionally similar to the vertebrate midbrain (Klein and Barron 2016). There is also evidence, mainly from bees, of working memory and of forms of associative learning that seem to be facilitated by conscious experience in humans (Birch 2022). These lines of evidence push us to take seriously the possibility of sentience in insects.

When we turn our attention to molluscs other than the cephalopods (such as gastropod molluscs, including snails) and to other arthropods (such as arachnids, including spiders) we find a frustrating paucity of studies looking for markers of sentience. There is some evidence of prospective simulation in *Portia* spiders, suggesting experiments aimed at exploring sentience in arachnids would be worthwhile (Cross et al. 2020). But we currently lack the sort of evidence base that would be needed to inform discussions about proportionality. These animals, along with nematode worms and insect larvae, are investigation priorities.

The growing interest in farming invertebrates as protein sources adds urgency to all these debates. We are seeing not just the growth of aquaculture, but also the rapid expansion of the insect farming industry. A plausibly proportionate response would be to set up a licensing scheme that requires companies farming sentience candidates or investigation priorities to obtain a licence, where a licence can only be obtained once clear, enforceable welfare standards exist. While I would be in favour of stronger action, a licensing scheme seems more likely to command consensus support than a ban.

However, in the case of octopus farming, a ban does have the potential to command consensus. Octopus farming cannot realistically meet reasonable expectations regarding animal welfare, so it would be proportionate to ban it, as I and 99 other experts have argued in a recent letter (Jacquet et al. 2024).

There are ongoing campaigns to require electrical stunning when shrimps are slaughtered, but we need more evidence to be confident that electrical stunning is effective and to get the parameters right. In the meantime, it is already clear that we should not use live boiling as a slaughter method for any decapod. This method needlessly risks extending and intensifying the suffering of these animals.

12. The emerging AI sentience debate. AI is developing at pace, and we must not be complacent about the risks of developing new kinds of sentient being (Butlin et al. 2023). At the same time, we must be alert to the risk of millions of users mistakenly attributing sentience to chatbots powered by large language models (LLMs). Chatbots create a powerful illusion of being your friend, assistant or romantic partner, leading more than half of users to believe there is some chance they are sentient (Colombatto et al. 2025). This illusion will rapidly become more and more persuasive, potentially distracting attention away from animals in which sentience is far more likely.

Why do I say “illusion”? Because, even though the inner workings of LLMs are substantially opaque, we know that the appearance of a persisting interlocutor, present throughout an extended conversation, is illusory. This is because every step in your conversation is processed separately. Each step can be routed to a different implementation of the LLM running in a different data center without any weakening of the illusion. It may be that one step in the conversation is processed in Virginia, the next in Vancouver, the next in Texas. All that persists across the discrete processing events is the context window containing the system prompt and conversation history. Put simply, *no one is there*. There is no sub-network anywhere in the world dedicated to your conversation. There is no friend, no assistant, no partner. But the illusion can be staggeringly, alarmingly convincing, especially when the chatbot presents with a human face and a human voice.

In the coming years, questions of AI sentience will only become more pressing and the demand for reliable tests will become ever more urgent. Yet in attempting to infer sentience in AI, we face the *gaming problem*: when an artificial agent can intelligently draw upon huge amounts of human-generated training data, it is well placed to game our criteria for sentience. Gaming occurs when systems mimic human behaviours that are likely to persuade human users of their sentience without possessing the underlying capacity. Criteria lose their usefulness once gamed. I liken this phenomenon to greenwashing: companies that know the criteria for eco-friendliness can game those criteria so well that immaculately ticking off all the boxes can become evidence *against* genuine eco-friendliness. When this happens, we need new criteria. What chatbots achieve is a kind of “sentience-washing”. No intentional deception is needed for gaming; it can happen in service of mundane objectives such as maximizing user-satisfaction or maximizing interaction time.

Three other pathways to artificial sentience candidates, though much less discussed, strike me as more credible than the chatbot case. The first involves emulating the brains

of biological sentience candidates such as insects, neuron by neuron, based on their connectomes. The resulting virtual brains are sentience candidates, in my view, if they display the same pattern of behavioural markers that we take as sufficient for sentience candidature in the biological original. The second path involves evolving artificial agents that converge, without gaming, on similar patterns of behavioural markers to biological sentience candidates. The third involves deliberately implementing a minimal version of a computational architecture (such as a global workspace) that has been credibly linked to sentience in humans.

All three pathways present ways in which we might come to create an artificial sentience candidate that does *not* display impressive feats of intelligence. We need to be mindful of the possibility of decouplings of sentience from intelligence in AI, just as in the animal case. This often gets forgotten in the discussion of chatbots.

Given the rate at which AI is developing, and the risks associated with artificial sentience taking us by surprise, we should apply the *run-ahead principle*: at any given time, measures to regulate the development of sentient AI should run ahead of what would be proportionate to the risks posed by current technology, considering also the risks posed by credible future trajectories.

The run-ahead principle may justify strong regulatory action. A moratorium, such as that proposed by Thomas Metzinger (2021), should be on the table, but has arguably become practically impossible—frankly, the moment has probably gone. Meanwhile, Susan Schneider’s (2020) more moderate alternative—involving regular testing to monitor the sentience of our AI creations—is currently unfeasible, given the absence of tests that can be applied to systems with high potential for gaming our criteria. A third approach involves oversight by means of sector-wide codes of good practice and licensing schemes. This path would require a greater level of transparency than we have seen from the AI industry to date.

The overarching imperative is to have a democratic debate about these questions, so as to prepare for the upheaval of human lives that artificial sentience candidates will precipitate if and when they arrive. In so doing, however, we must be careful not to take our eye off the quintillions of living sentience candidates that need our attention and compassion now.

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